



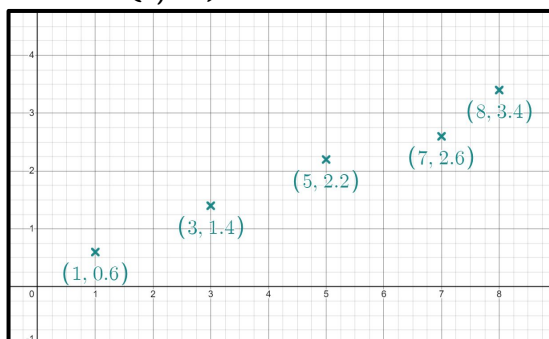
Plotting coordinates generated from a rule using technology

Task A: Plotting coordinates in Desmos

- On your graphing software press the  button if visible.
Change the step on your x axis to 3. Change the step on your y axis to 4
What integer values do your axes range from and to?
For most display screens the x axis will range from -9 to 9 and the y axis from -4 to 4
- Move your axis until you can see 28 on the y axis. What is the next value visible below 28? 24
- Press the  button. Describe what this does to your axes.
It centres the screen around the origin. It also changes the axis so they have the same scale (although the steps are still different)
- Plot the coordinates (2,4) and (2,6). Plot a third coordinate so it is in the middle of these coordinates. What coordinate did you plot? (2,5)

Task B: Identifying coordinates that do not follow a rule

- Plot each set of coordinates in turn. Identify the coordinate which is the odd one out.
 - (0,-4) (4,8) (1,-1) **(2,6)** (-2,-10)
 - (0,7)** (4,4) (10,-2) (-1,9) (2,6)
 - (8,-1.4) (5,-1.85) **(-4, 3.2)** (12,-0.8) (-9,-3.95)
 - (11,2.9) **(1, $\frac{3}{10}$)** (-9,-1.1) (2, $1\frac{1}{10}$) (-5, $-\frac{3}{10}$)
- Izzy has plotted the coordinates below.
 - Which one do you think she has plotted incorrectly? How do you know?
(7,2.6) because it doesn't sit on a straight line with the other 4 points



- Recreate Izzy's graph
 - make the incorrect coordinate draggable.
 - Drag the coordinate so it is in line with the rest and on a coordinate where both values are integers.
 - What coordinate have you now plotted?
(7,3) or (2,1) There are other possibilities not visible on this graph



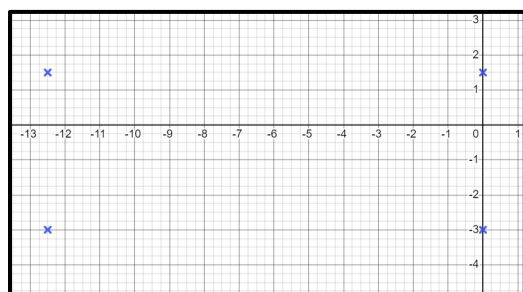
Task C: Creating shapes in desmos

For each set of coordinates, find the fourth coordinate to make a square

- a) $(0,0)$ $(8,-2)$ $(10,6)$ $(2,8)$
- b) $(0,0)$ $(4,5)$ $(-1,9)$ $(-5,4)$
- c) $(0,0)$ $(7,-3)$ $(3,7)$ $(10,4)$
- d) $(2,3)$ $(-1,-4)$ $(4,-2)$ $(-3,1)$
- e) $(-10,0.5)$ $(-8,1.5)$ $(-9,-1.5)$ $(-7,-0.5)$

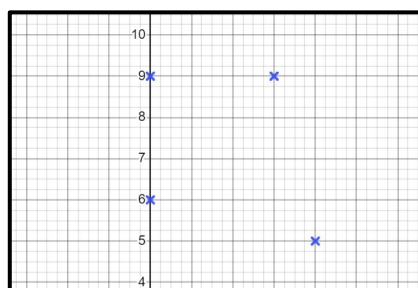
2) Plot each set of coordinates to determine the type of quadrilateral they make.

- a) $(-12.5,-3)$ $(-12.5, 1.5)$ $(0,-3)$ $(0,1.5)$



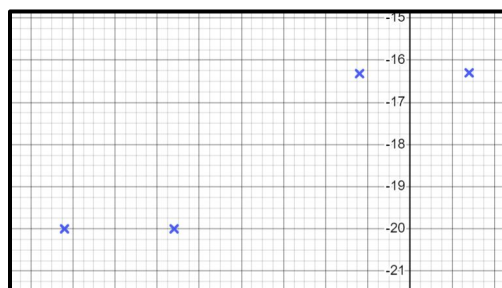
Rectangle

- c) $(0,9)$ $(3,9)$ $(0,6)$ $(4,5)$



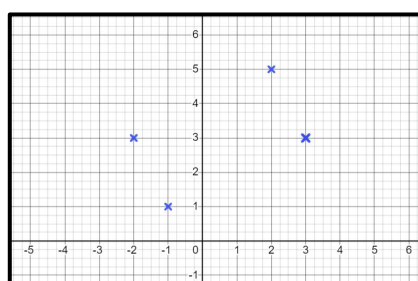
Kite

- b) $(-8.2,-20)$ $(1.4,-16.3)$ $(-1.2,-16.3)$ $(-5.6,-20)$



Parallelogram

- d) $(-2,3)$ $(3,3)$ $(2,5)$ $(-1,1)$



Rectangle

